

Logarithms - 0.1.5

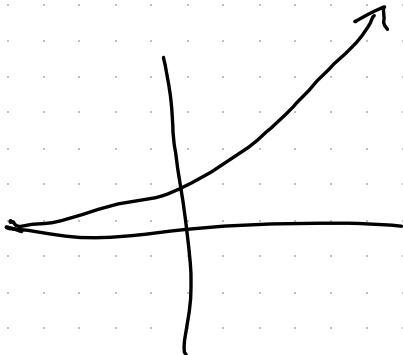
Recall: we defined a^x for all real x and $a > 0$

This is a function with domain $(-\infty, \infty)$
and range $(0, \infty)$

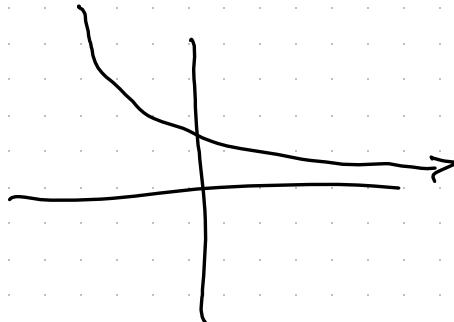
$a > 1$: increasing

$a < 1$: decreasing

$\Rightarrow a^x$ is one-to-one,
so invertible



$a > 1$

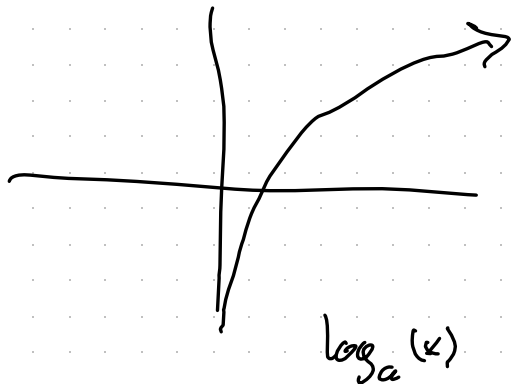


$a < 1$

Definition: The inverse of the function a^x is the logarithm with base a , denoted by $\log_a(x)$.

That is, $\log_a(x) = y$ if and only if $a^y = x$.

$\log_a(x)$ has domain $(0, \infty)$ and range $(-\infty, \infty)$.



Examples: $\log_2(8) = 3$
because $2^3 = 8$.

$\log_{10}(\frac{1}{100}) = -2$ because
 $10^{-2} = \frac{1}{10^2} = \frac{1}{100}$.

Properties: let $a, b, c > 0$, r be a real #

$$1. a^{\log_a(bc)} = bc = a^{\log_a(b)} a^{\log_a(c)} = a^{\log_a(b) + \log_a(c)}$$

$$\Rightarrow \log_a(bc) = \log_a(b) + \log_a(c)$$

$$2. \log_a(b/c) = \log_a(b) - \log_a(c)$$

$$3. \log_a(b^r) = r \log_a(b)$$

Example: Solve $\log_{10}(\sqrt{x}) + \log_{10}(x) = 2$ for x .

$$\log_{10}(x^{\frac{1}{2}}) + \log_{10}(x) = \frac{1}{2} \log_{10}(x) + \log_{10}(x) = \frac{3}{2} \log_{10}(x)$$

$$\log_{10}(x) = \frac{4}{3} \Leftrightarrow x = 10^{4/3}$$

We can change bases:

$$- a^x = (b^{\log_b(a)})^x = b^{x \log_b(a)}$$

$$- \log_a(x) = \log_a(b^{\log_b(x)}) = \log_b(x) \cdot \log_a(b)$$

$$\Leftrightarrow \log_b(x) = \frac{\log_a(x)}{\log_a(b)}$$

Notation: since e is special, $\log_e(x)$ gets a special name: $\ln(x)$ "natural log"